Creating Custom Equity Indexes

FIN 684 - Investment Theory / Advanced Corporate Finance (Prof. Sury), Project I

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Overview

The goal of this project was to build and analyze three custom equity indexes (equal-weighted, value-weighted, and price-weighted) using historical U.S. stock return data from CRSP. We compared these indexes against popular ETF benchmarks (SPY, DIA, and QQQ) to see how different weighting methods affect performance and correlations. Our code prompts the user for a desired start date (2000-2024), end date (2000-2024), and the number of stocks in the index (1-2000). Our example index construction focused on the top 100 stocks by lagged market cap from 2000 to 2024. We built monthly total-return indexes for each weighting method while keeping realistic assumptions to avoid biases, and we visualized the results through a time-series plot and a correlation matrix of monthly log returns.

Methodology

We started by pulling monthly stock data from the CRSP database, applying filters during the query to include only U.S. common shares trading on major exchanges (NYSE, AMEX, and NASDAQ). From the price and shares-outstanding data, we calculated market capitalization for each stock. Our index construction implements monthly reconstitution and rebalancing under a systematic process. At each month-end t, we ranked all eligible securities by market capitalization using data from t-1 and selected the top 100 stocks to form our index for month t. Three weighting schemes were applied: equal-weighted, where each constituent receives a weight of 1/N; value-weighted, where weights are proportional to market capitalization; and price-weighted, where weights are proportional to share price. Index returns are the weighted average of constituent stock returns, with all indexes initialized at a base level of 100. The same logic was used for the benchmark ETFs.

Each month, we selected the top 100 stocks based on market capitalization from the previous month. This avoids look-ahead bias because we only used information that would have been available at the time of portfolio formation. We also pulled delisting returns to compute effective total returns that properly account for failed companies, combining the regular return (ret) with the delisting return (dlret) using the formula (1 + ret) x (1 + dlret) - 1. This helped us deal with survivorship bias, since delisted stocks are often missing from typical backtests. When delisting returns were missing, we assumed them to be 0, which is a conservative but fair assumption. Mitigating look-ahead and survivorship bias is crucial because these biases make backtests appear far more profitable than what investors could have actually achieved in real time. Without proper bias controls, a strategy might show exceptional historical returns that are essentially impossible to replicate in practice.

We implemented several data standardization steps to ensure accuracy and consistency. Price values are converted to absolute values to handle CRSP's negative-price convention for certain quote types, shares outstanding are properly scaled, and the data is processed in annual chunks to manage memory efficiently when handling over 1.2 million records. It is worth noting that we originally considered using IWM and VTI as benchmarks, but those did not have full history going back to 2000. Instead, we used SPY (tracks the S&P 500), QQQ (tracks the NASDAQ-100), and DIA (tracks the Dow Jones), which had complete data coverage for the entire period. As stated above, our example specification chose the top 100 stocks with a start date of 2000 and an end date of 2024. The time-series plot and correlation-matrix heatmap are shown in the exhibits below.

Performance Results

Our three indexes showed very different results over the 25-year period (2000-2024). The price-weighted index achieved an exceptional total return of 1,311.9% (11.2% annualized), far exceeding both other custom indexes and all benchmarks. The value-weighted index generated a 504.6% total return (7.5% annualized), while the equal-weighted index produced 438.3% (7.0% annualized). These represent total growth from a starting value of 100, meaning a $100 investment in the price-weighted index in 2000 would have grown to about $1,411.90 by the end of 2024. Among the benchmark ETFs, DIA returned 619.4%, SPY returned 569.4%, and QQQ returned 462% over the same period. Compared to these benchmarks, our price-weighted index was the only one to outperform across all measures.

Interestingly, our value-weighted index (504.6%) underperformed SPY (569.4%) despite using similar market-cap weighting. The difference likely reflects our use of 100 stocks versus the S&P 500's 500 stocks, suggesting that mid-cap names in the S&P 500 added value over this period. Our equal-weighted index's lower return (438.3%), compared to both the value-weighted index and the benchmarks, indicates that giving equal weight to the smallest stocks within our top 100 did not enhance returns.

Correlation and Co-movement

The correlation analysis reveals how closely our indexes tracked established benchmarks. Our equal-weighted index shows near-perfect correlation with SPY (0.99), which seems counterintuitive given their different weighting schemes. This high correlation implies that among the largest 100 stocks, weighting differences may have minimal impact on directional co-movement even if they drive differences in total return. The value-weighted index also correlates highly with SPY (0.98) and DIA (0.93), confirming that these indexes capture similar market movements. The price-weighted index, by contrast, stands apart with much lower correlations to the benchmarks (0.33 to 0.61), showing that it followed a very different pattern.

Among the three ETFs, the price-weighted index was closest to DIA (0.61), which is itself a price-weighted index.

This divergence is likely tied to the higher volatility of the price-weighted index, which concentrates exposure in fewer, high-priced stocks that can swing more dramatically. In contrast, the equal- and value-weighted indexes benefit from more diversified exposure and therefore exhibit more stable return patterns. Notably, the price-weighted index began to significantly outperform after 2016 and again after 2020. This surge is likely due to the dominance of high-priced leaders (such as the Magnificent 7 and Berkshire Hathaway) that avoided stock splits and experienced exceptional price growth. Because price-weighting gives greater weight to stocks with higher prices regardless of fundamentals, it concentrates the portfolio in a few high-performing names. This momentum-like effect magnified returns but also reduced diversification. While this index delivered the highest returns, the approach has practical limitations: it can overweight companies purely based on share price while ignoring fundamentals like market capitalization or profitability.

Discussion

Looking at the time-series chart, all indexes moved together during major market events like the 2008 financial crisis and the COVID-19 drawdown and recovery. The recovery paths of the equal- and value-weighted indexes were similar throughout the period, while the price-weighted index consistently tracked higher and eventually diverged sharply from the others. The surprising underperformance of equal-weighting versus value-weighting runs against common expectations about a small-cap premium. Within our top 100 stocks, giving equal weight to the smallest large-caps actually hurt returns. This suggests that among mega-cap stocks size does matter, as the very largest companies outperformed those ranked 50 to 100 by market cap.

The correlation matrix also speaks to diversification. Despite differences in returns, our equal- and value-weighted indexes correlate strongly with the benchmarks, meaning they offer little diversification benefit to an investor who already owns SPY or a similar fund. The price-weighted index's relatively low correlations across the board suggest it could provide diversification benefits when held alongside one of the benchmarks, though this comes with the trade-off of reduced diversification within the index itself.

Reflections on Challenges

Beyond the technical steps, several data challenges highlighted how fragile empirical finance can be if data issues are not handled carefully. CRSP's negative-price convention and missing delisting returns reminded us that small data quirks can meaningfully distort results if overlooked. Even simple choices, like assuming a 0% delisting return when none is provided, can change total-return calculations. Handling memory limits also underscored the importance of writing efficient, reproducible code when working with millions of records. Overall, building credible indexes is not only about coding the formulas correctly, but also about managing messy data, making transparent assumptions, and understanding the trade-offs behind them.

Exhibit A: Index Levels (Custom Indexes vs. ETF Benchmarks), 2000-2024

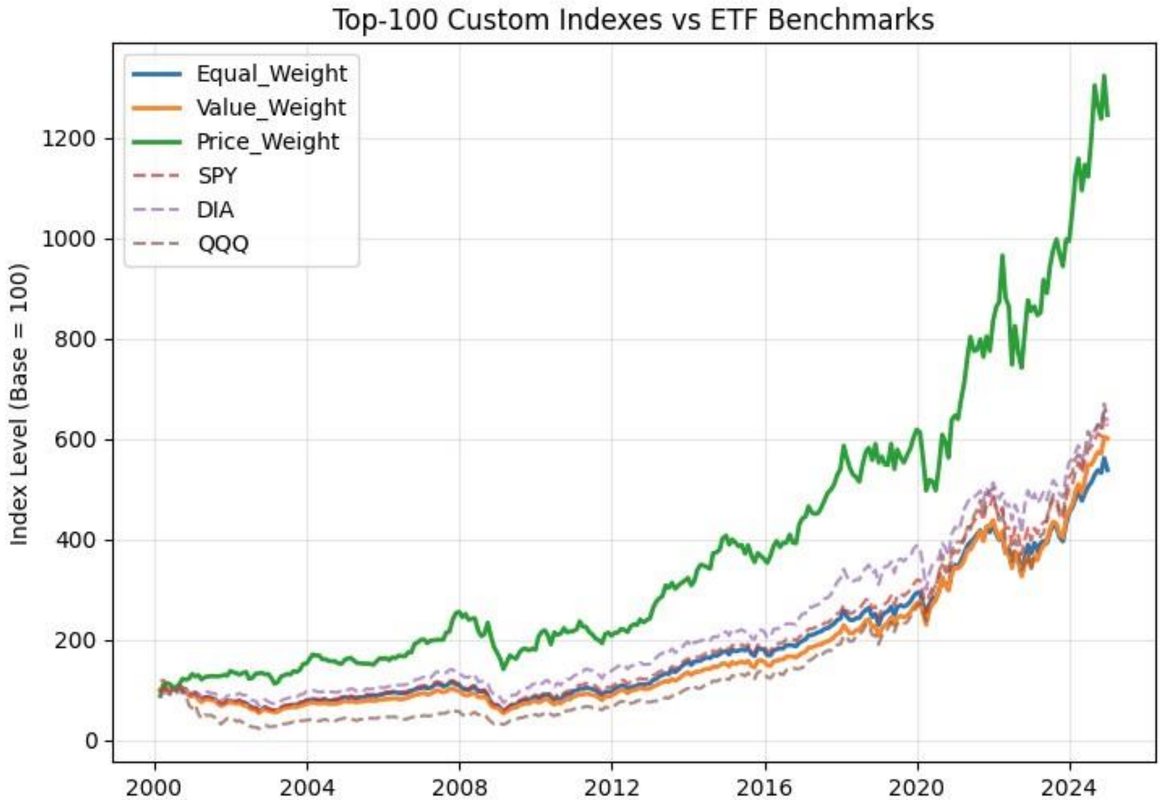

Exhibit B: Correlation Matrix of Monthly Log Returns